SOLSTICE: TASK 1

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Date: ~~Probably~~ Most definitely too late[but hopefully good quality :)]

§1 Understanding the Setup

1.1 Notation

To prevent ANY confusion later on, I just want to clarify the notation I would like to use for the rest of this analysis:

- For referring to random variables I will use capital letters like X, Y or $\overline{W}$.
- If the quantity I am referring is a vector, I will use vector notation. For example, $X^n \equiv \overline{X} = (x_1, x_2, \ldots, x_n)$. Please do not confuse the bar on the top for averages or expectation values.
- I will use angled brackets $\langle X \rangle$, or $\mathbb{E}[X]$ to denote the expectation of some (say) random variable X.
- **Truth** $q(x)$: I will reuse **q** without changing it for all the true distributions.
- **Model** $p(x|\overline{w})$
- **Prior** $\varphi_1(\overline{w})$: The subscript 1 denotes that this is the initial prior. If we do 1 Bayesian update we get $\varphi_2(\overline{w}|\overline{x})$, where $\overline{x}$ is the new information we have observed. Sometimes I might omit the $\overline{x}$ part and just write $\varphi_2(\overline{w})$ because I am too lazy to type so much each time.

1.2 Analysis for $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$

- **Truth**: $q(x, y)$
- **Model**: $p(x, y|\overline{w})$
- **Prior**: $\varphi_1(\overline{w})$
- **Posterior**: $\varphi_2(\overline{w}|(x_0, y_0)) = \dfrac{p(x_0, y_0|\overline{w})\varphi_1(\overline{w})}{\int p(x_0, y_0|\overline{w})\varphi_1(\overline{w})d\overline{w}}$
- **Expectation Value** of a function $f(\overline{w}) : \mathbb{R}^n \to \mathbb{R}$ over the posterior:

$$\mathbb{E}[f(\overline{w})] = \int f(\overline{w})\varphi_2(\overline{w})d\overline{w}$$

$$\mathbb{E}[f^2(\overline{w})] = \int f^2(\overline{w})\varphi_2(\overline{w})d\overline{w}$$

$$\mathbb{V}[f(\overline{w})] = \mathbb{E}[f^2(\overline{w})] - \mathbb{E}[f(\overline{w})]^2$$

- **Predictive Density Function**:

$$\hat{p}(x, y) = \int p(x, y|\overline{w})\varphi_2(\overline{w})d\overline{w}$$

NOTE:

1. If we put $\varphi_1(\overline{w})$ instead of $\varphi_2(\overline{w})$ in the above equation, the function evaluated by that integral would be our initial guess for $q(x, y)$.
2. After every Bayesian step, $\varphi_i(\overline{w})$ is the only part of the above integral that changes. The model: $p(x, y|\overline{w})$ stays the same.
3. The function inside the integral is the joint PDF of x, y and $\{w_i\}_{i=1}^{n_w}$ where w_n is the dimensionality of the weight parameter's vector space. The marginal of that PDF w.r.t. x, y is the predictive density function
4. $\hat{p}(x, y) = \mathbb{E}_{\varphi_2}[p(x, y|\overline{w})]$

1.3 A special case when $q(x)$ is known

We still observe the same $\mathcal{D}_n = \{(x_i, y_i)\}_{i=1}^n$. We make a very small tweak to our setup if we already know $q(x)$. Knowing $q(x)$ heavily restricts our model choices, since it's marginal w.r.t. x must give $q(x)$. So we are selecting a conditional PDF $p(y|x, \overline{w})$ and not $p(x, y|\overline{w})$
Because of this, some expressions change/simplify slightly:

- **Posterior:**

$$\varphi_2(\overline{w}|(x_0, y_0)) = \frac{p(x_0, y_0|\overline{w})\varphi_1(\overline{w})}{\int p(x_0, y_0|\overline{w})\varphi_1(\overline{w})d\overline{w}}$$

$$\varphi_2(\overline{w}|(x_0, y_0)) = \frac{p(y_0|x_0, \overline{w})p(x_0|\overline{w})\varphi_1(\overline{w})}{\int p(y_0|x_0, \overline{w})p(x_0|\overline{w})\varphi_1(\overline{w})d\overline{w}}$$

$$\varphi_2(\overline{w}|(x_0, y_0)) = \frac{p(y_0|x_0, \overline{w})q(x_0)\varphi_1(\overline{w})}{\int p(y_0|x_0, \overline{w})q(x_0)\varphi_1(\overline{w})d\overline{w}}$$

We can cancel the $q(x_0)$ since it is a constant, to get:

$$\varphi_2(\overline{w}|(x_0, y_0)) = \frac{p(y_0|x_0, \overline{w})\varphi_1(\overline{w})}{\int p(y_0|x_0, \overline{w})\varphi_1(\overline{w})d\overline{w}}$$

- **Predictive Density Function:**

$$\hat{p}(x, y) = \int p(x, y|\overline{w})\varphi_2(\overline{w})d\overline{w}$$

$$\hat{p}(y|x) = \frac{\hat{p}(x, y)}{\int \hat{p}(x, y)dy}$$

$$\hat{p}(y|x) = \frac{\int p(x, y|\overline{w})\varphi_2(\overline{w})d\overline{w}}{\int\int p(x, y|\overline{w})\varphi_2(\overline{w})(d\overline{w})dy}$$

$$\hat{p}(y|x) = \frac{\int p(y|x, \overline{w})p(x|\overline{w})\varphi_2(\overline{w})d\overline{w}}{\int\int p(y|x, \overline{w})p(x|\overline{w}))\varphi_2(\overline{w})(d\overline{w})dy}$$

$$\hat{p}(y|x) = \frac{\int p(y|x, \overline{w})q(x)\varphi_2(\overline{w})d\overline{w}}{\int\int p(y|x, \overline{w})q(x)\varphi_2(\overline{w})(d\overline{w})dy}$$

Since $q(x)$ is a "constant" in the denominator's integral, we can cancel the $q(x)'s$.

$$\hat{p}(y|x) = \frac{\int p(y|x, \overline{w})\varphi_2(\overline{w})d\overline{w}}{\int\int p(y|x, \overline{w})\varphi_2(\overline{w})(d\overline{w})dy}$$

The denominator beautifully evaluates to be 1, and so we get:

$$\hat{p}(y|x) = \int p(y|x, \overline{w})\varphi_2(\overline{w})d\overline{w}$$

These two results of the posterior and predictive make it seem like the model is really the conditional PDF and not the joint. While I guess you could draw that conclusion, I like to think of the model as always being a joint PDF because it helps me understand the method of analysis better.

End of Q1

§2 Gradient Descent v/s Bayesian Updating

Given Situation

- We observe: $\mathcal{D}_3 = \{(0,1),(1,2),(2,3)\}$
- Goal: We want a "model" to tell us what y will be for a given x value

2.1 Gradient Descent Approach

- **Step 1: Loss function:-**

$$\mathcal{L}(m,c) = \frac{1}{3}\sum_{i=1}^{3}(y_i - mx_i - c)^2$$

- **Step 2: Gradients:-**

$\boxed{\text{We are told to initialize our slope and intercept with } \overline{w} = [0,0]}$

$$\nabla\mathcal{L} = \left[\ \left(\frac{\partial\mathcal{L}}{\partial m}\right)_c, \left(\frac{\partial\mathcal{L}}{\partial c}\right)_m\right]$$

$$\nabla\mathcal{L} = \left[\ \left(-\frac{2}{3}\sum_{i=1}^{3}\{(y_i - mx_i - c)(x_i)\}\right), \left(-\frac{2}{3}\sum_{i=1}^{3}(y_i - mx_i - c)\right)\right]$$

$$\nabla\mathcal{L} = \left[\ \left(-\frac{2}{3}\sum_{i=1}^{3}\{(y_i)(x_i)\}\right), \left(-\frac{2}{3}\sum_{i=1}^{3}(y_i)\right)\right]$$

$$\nabla\mathcal{L} = \left(-\frac{16}{3}, -2\right)$$

- **Step 3: Update Weights:-**

$\boxed{\text{Given learning rate } \eta = 0.01}$

$\overline{w_1} = \overline{w_0} - \eta(\nabla\mathcal{L})$

$\overline{w_1} = -\eta(\nabla\mathcal{L})$

$\overline{w_1} = -0.01\left(-\frac{16}{3}, -2\right)$

$\boxed{\overline{w_1} = (0.053, 0.02)}$

2.2 The Bayesian Approach

- **Model:** $p(x, y|\overline{w}) = e^{-\frac{1}{2}(y-mx-c)^2}$, and each (x_i, y_i) is independent of (x_j, y_j) $\forall i \neq j$
- **Prior:** $\varphi_1(\overline{w}) = e^{-\frac{1}{2}(m^2+c^2)}$
- **Truth:** $q(x)$ is a uniform distribution with every equal sized range on the line $y = x+1$ having equal probability, i.e. $q(x) = \delta(y - x - 1)$

  The model is also often called the likelihood term $\left(\text{though I would argue the likelihood term is} \dfrac{p(x|\overline{w})}{Z(X)}\right)$

- **Posterior:**

$$\varphi_2(\overline{w}|\mathcal{D}_3) \equiv \varphi_2(\overline{w}) = \frac{\prod_{i=1}^{3} p(x_i, y_i|\overline{w})\varphi_1(\overline{w})}{Z(X^3)}$$

$$\varphi_2(\overline{w}) \propto \exp\left(-\frac{1}{2}\left(m^2 + c^2 + (1-c)^2 + (2-m-c)^2 + (3-2m-c)^2\right)\right)$$

- **Predictive Density:**

$$\hat{p}(x, y|\mathcal{D}_3) = \int p(x, y|\overline{w})\varphi_2(\overline{w})d\overline{w}$$

$$\hat{p}(x, y|\mathcal{D}_3) \propto \int \exp\left(-\frac{1}{2}\left(m^2 + c^2 + (1-c)^2 + (2-m-c)^2 + (3-2m-c)^2 + (y-mx-c)^2\right)\right) dm\ dc$$

  This can then be solved using epic vectors and matrices. Basically we rotate and shift the conic (the exponent of the exp) in (m, c) into an ellipse centered at the origin so that we can use the gaussian integrals to get the required answers.

My Interpretation of the difference between GD and The Bayesian Approach

Goal: I have data points $\{(x_i, y_i)\}_{i=1}^n$, and I want to find a relationship between them

- **Gradient Descent/Classical Methods**: Throw these points on a 2-D plane, and the relationship we are looking for is a (best-fit) function.
- **Bayesian Approach:** These points are there on a 2-D plane, but they are there because above the plane there is a HUGE blanket $\rightarrow$ the joint PDF of (X, Y), from which these points got sampled! This method accounts better for the randomness! If some random observation (89567234, 12389567234) comes in the dataset which is far away from the function predicted by gradient descent, this is not too surprising for Bayesian statistician since that point just had a very low density around it, but not impossible!

End of Q2

§3 Understanding MAP and MLE

I will update this section after understanding these better. The calculation parts are straightforward, but understanding and using these estimators is something I haven't fully wrapped my head around yet.

End of Q3

§4 Losses, Errors and Empirical Calculations

4.1 Entropy

Setup: There is a random variable X, its true distribution is $q(x)$. The priors are $\varphi_i(\overline{w})$ and the model is $p(x|\overline{w})$. x_i represents the ith sample from X. X_i is just for our understanding/convenience. It is identical to X in every way. Given $\overline{W}$, all the $X_i's$ are independent.

Entropy: $S = -\int q(x) \ln q(x) dx$,

This quantity is the expected surprisal/information in the distribution $q(x)$.

Empirical Entropy: $S_n = -\frac{1}{n}\sum_{i=1}^{n} \ln q(x_i)$

This is the "experimental" entropy. We observe a finite number of samples from $q(x)$, and the moment we observe those n samples, the value of S_n is determined.

4.1.1 Expectation Values

- $\mathbb{E}[\mathbf{S_n}]$: I think of this in three ways:

  1. **Intuitive Argument:** By LLN, the expectation value of S_n is basically calculating S_n billions of times and then averaging them out. If I want to calculate S_n many times, it immediately implies that I must take as many batches of n samples of X too. Taking many S_n's and then averaging is therefore equivalent to letting $n \to \infty$ in the formula for S_n. Now, since each x_i comes out of $q(x)$, by LLN :- $\lim_{n\to\infty} S_n = \mathbb{E}[S_n] = S$!

  2. S_n **is a random variable, a function on** X^n:

     $$\mathbb{E}[S_n] = \int S_n \left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1 dx_2 \dots dx_n)$$

     $$\mathbb{E}[S_n] = \int \left(-\frac{1}{n}\sum_{i=1}^{n} \ln q(x_i)\right)\left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1 dx_2 \dots dx_n)$$

     $$\mathbb{E}[S_n] = -\frac{1}{n}\int \left(\sum_{i=1}^{n} \ln q(x_i)\right)\left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1 dx_2 \dots dx_n)$$

     $$\mathbb{E}[S_n] = -\frac{1}{n}\sum_{i=1}^{n}\left(\int \ln q(x_i)\left(q(x_1)q(x_2)\cdots q(x_i)\dots q(x_n)\right)(dx_1 dx_2 \dots dx_i \dots dx_n)\right)$$

     The un-highlighted parts all integrate to one (the above integral is actually a multi-dimensional integral which I'm representing with just one $\int$ symbol instead of $\iiint \dots \int_{X^n}$ because it looks cleaner. Thus the above expression simplifies to:

     $$\mathbb{E}[S_n] = -\frac{1}{n}\sum_{i=1}^{n}\left(\int \ln q(x_i) q(x_i) dx_i\right)$$

     The highlighted integral is $-S$, therefore we prove:

     $\mathbb{E}[S_n] = S$! [that's an exclamation not a factorial :)]

  3. S_n **random variable, a SUM of *independent* random variables** $-\ln(X_i)$**:** This gives the answer immediately :-

$$\mathbb{E}[S_n] = \frac{1}{n}\sum_{i=1}^{n}\mathbb{E}[-\ln(q(x_i))]$$

$$\mathbb{E}[S_n] = \frac{1}{n}\sum_{i=1}^{n}\left(-\int q(x_i)\ln q(x_i)dx_i\right)$$

$\mathbb{E}[S_n] = S$:)

Although finishing the proof in 3 lines is not as satisfying as arguing through LLN as in method (1), or writing a bunch of cool letters and symbols as in method (2), it is quite elegant at the end of the day.

Method (3) is arguably a more "mature" version of method (2). It's often useful to think in terms of composing into a bunch of independent random variables.

- $\mathbb{E}[\mathbf{S_n^2}]$:

Unfortunately, I couldn't think of any solution other than to write a bunch of integrals, though doing that is quite therapeutic in it's own sense.

$$\mathbb{E}[S_n^2] = \int S_n^2\left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1dx_2\dots dx_n)$$

$$\mathbb{E}[S_n^2] = \int \frac{1}{n^2}\left(\sum_{i=1}^{n}\ln q(x_i)\right)^2\left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1dx_2\dots dx_n)$$

$$\mathbb{E}[S_n^2] = \frac{1}{n^2}\int\left(\sum_{i=1}^{n}\ln q(x_i)\right)^2\left(q(x_1)q(x_2)\dots q(x_n)\right)(dx_1dx_2\dots dx_n)$$

Define: $I_{\alpha\beta} = \int \ln q(x_\alpha)\ln q(x_\beta)\left[q(x_1)\dots q(x_\alpha)\dots q(x_\beta)\dots q(x_n)\right](dx_1\dots dx_\alpha\dots dx_\beta\dots dx_n)$

Case 1: $\alpha \neq \beta$**:**

$I_{\alpha\beta} = \underbrace{\int q(x_\beta)\ln q(x_\alpha)dx_\alpha}_{-S}\cdot\underbrace{\int q(x_\beta)\ln q(x_\beta)dx_\beta}_{-S}$; All the other integrals factor out and integrate to 1.

$I_{\alpha\beta} = S^2$

Case 2: $\alpha = \beta$**:**

$I_{\alpha\alpha} = \int \ln q(x_\alpha)\ln q(x_\alpha)q(x_\alpha)dx_\alpha$; All the other integrals integrate to 1.

$I_{\alpha\alpha} = I_0 = \int q(x)\left(\ln q(x)\right)^2 dx$

$$\mathbb{E}[S_n^2] = \frac{1}{n^2}\left(nI_0 + 2\ {}^nC_2\ S^2\right)$$

$$\mathbb{E}[S_n^2] = \frac{I_0}{n} + \frac{(n-1)}{n}S^2$$

$$\mathbb{E}[S_n^2] - S^2 = \frac{I_0}{n} - \frac{S^2}{n}$$

$$\mathbb{E}[S_n^2] - \mathbb{E}[S_n]^2 = \frac{1}{n}\left(\int q(x)\left(\ln q(x)\right)^2 dx - S^2\right)$$

$$\boxed{\mathbb{V}[S_n] = \frac{1}{n}\left(\int q(x)\left(\ln q(x)\right)^2 dx - S^2\right)}$$

4.1.2 Intuitive Interpretation of Entropy

- **View 1:** We can think of it as the expected amount of surprisal/information in a distribution.
- **View 2:** We can think of entropy as uncertainty in a distribution too! As in, the larger the entropy, the larger are the "possible" set of X values.

  I feel like view 2 makes sense based on the entropy I know from physics, and view 1 makes sense to me based on the way we define it here. But why are they equivalent? I spent quite a while thinking about this, and I haven't come to satisfying conclusion yet. Both make sense in their own rights, but I haven't yet thought of a way to tie them together.

4.2 Generalization and Training Loss

$$G_n = -\int q(x) \ln \hat{p}(x|X^n)\ dx$$

$$T_n = -\frac{1}{n}\sum_{i=1}^{n} \ln \hat{p}(x_i|X^n)\ dx$$

By qualitative observation we can say that G_n is a better metric to compare models since T_n is evaluated on the observed data, whilst G_n is evaluated on all possible data. This means is that if n is very large, T_n might end up being small, but G_n could still be decent.

4.2.1 Relationship between $\mathbf{G_n}$ and S

Intuitively, S is the ideal value we want our predictive to achieve when put into G_n. So $G_n - S$ measures our models deviation. After seeing many data points, we want $\hat{p}(x) \to q(x)$, and thus $G_n \to S$. But is $G_n > S$, or $G_n < S$? The Calculus of Variations (thank you Sudhanva) helps us mathematically ground this intuition, and answer the inequality.

Consider the functional:

$Q[f] = -\int q(x) \ln f(x)\ dx$, where $f(x)$ is a valid PDF.

Let $\tilde{f}(x)$ be the stationary point of this functional. Thus consider $h(x) = \tilde{f}(x) + \epsilon g(x)$, where $g(x)$ is any function that satisfies $\int g(x)\ dx = 0$(ensures that h is also normalized).

$$Q[\tilde{f} + \epsilon g] = -\int q(x) \ln\left(\tilde{f}(x) + \epsilon g(x)\right)\ dx$$

$$\frac{\partial Q}{\partial \epsilon} = -\int q(x) \frac{g(x)}{\tilde{f}(x) + \epsilon g(x)}\ dx$$

Since $\tilde{f}(x)$ is a stationary point,

$$\left|\frac{\partial Q}{\partial \epsilon}\right|_{\epsilon=0} = -\int \left|q(x)\frac{g(x)}{\tilde{f}(x) + \epsilon g(x)}\right|_{\epsilon=0}\ dx \overset{set}{=} 0$$

We therefore get the following:

$$-\int \frac{q(x)}{\tilde{f}(x)} g(x)\ dx = 0, \forall g(x)\ s.t. \int g(x)dx = 0$$

This implies (along with normalization) that:

$$\boxed{\tilde{f}(x) = q(x)}$$

Using the second derivative w.r.t ϵ, we can prove that $q(x)$ minimizes the functional.

We therefore get that $G_n - S \geq 0$, and the intuition that $G_n - S$ is some kind of "error" turns out to be supported by the math! Simply lovely.

$$\boxed{G_n - S \text{ is called the } \textbf{Kullback–Leibler(KL) divergence} \text{ of } \hat{p}(x) \text{ from } q(x)}$$

Another interpretation: If we take an observation x, then according to our predictive we get some surprise from it, and according to the truth we get some other amount of surprise from it. The expected delta in this surprise is the KL divergence.

4.3 Cross Validation Loss

$$C_n = -\frac{1}{n} \sum_{i=1}^{n} \ln \hat{p}(x_i | X^n \setminus X_i)$$

Note:

1. C_n unlike G_n, can be calculated in the real world, as it doesn't require knowledge of the true distribution $q(x)$. In fact, C_n is the point estimator for the value of G_n
2. Each $\hat{p}(x_i|X^n \setminus X_i)$ are DIFFERENT functions! Each of these predictives come from the same prior, but different likelihoods (but same model!)

<u>PODASIPs</u>

1. $\mathbb{E}[\mathbf{C_n}]$:

$$\mathbb{E}[C_n] = -\frac{1}{n} \int \left(\sum_{i=1}^{n} \ln \hat{p}(x_i|X^n \setminus X_i) \right) (q(x_1)q(x_2)\dots q(x_n))(dx_1 dx_2 \dots dx_n)$$

$$\mathbb{E}[C_n] = -\frac{1}{n} \sum_{i=1}^{n} \int \ln \hat{p}(x_i|X^n \setminus X_i)(q(x_1)q(x_2)\dots q(x_n))(dx_1 dx_2 \dots dx_n)$$

In each integral, 1 data point (corresponding to the value of i) is dropped. The limits for each integral include ALL the relevant coordinates of the remaining variables. Since all the variables are *i.i.d.*, these coordinates will be the same. Thus it immediately follows that all the n integrals, irrespective of the i value, will be equal! It also follows that it doesn't matter which of the n data-points I drop. Let's just say it's the nth one. Therefore:

$$\mathbb{E}[C_n] = \int \ln \hat{p}(x_i|X^{n-1})(q(x_1)q(x_2)\dots q(x_n))(dx_1 dx_2 \dots dx_n) = \underbrace{\int \ln \hat{p}(x_i|X^{n-1})(q(x_1)\dots q(x_{n-1}))(dx_1 \dots dx_{n-1})}_{\int q(x_n)dx_n = 1, \text{so I removed it.}}$$

$$\boxed{\textbf{Proved: } \mathbb{E}[C_n] = \mathbb{E}[G_{n-1}]}$$

2. Alternate Formulation of C_n?

$$-\frac{1}{n}\sum_{i=0}^{n}\mathbb{E}_{\overline{w}}[\ln p(x_i|\overline{w})] = -\frac{1}{n}\mathbb{E}_{\overline{w}}\left[\sum_{i=0}^{n}\ln p(x_i|\overline{w})\right]$$

$$-\frac{1}{n}\mathbb{E}_{\overline{w}}\left[\sum_{i=0}^{n}\ln p(x_i|\overline{w})\right] = -\frac{1}{n}\mathbb{E}_{\overline{w}}\left[\ln\left(\prod_{i=0}^{n}p(x_i|\overline{w})\right)\right] \neq C_n!$$

$$-\frac{1}{n}\mathbb{E}_{\overline{w}}\left[\ln\left(\prod_{i=0}^{n}p(x_i|\overline{w})\right)\right] \neq C_n!$$

I will update this section after understanding this better. I understand this conceptually, but I do not understand what the algebra will look like for the predictive after leaving one observation.

End of Q4

§5 Free ~~Energy~~ S Grade

$$Z(x_1, x_2, \ldots, x_n) \equiv Z(X^n) = \int \prod_{i=1}^{n} p(x_i|\overline{w})\varphi_1(\overline{w})\ d\overline{w}$$

As I mentioned in a note in Q1, this denominator is like our initial guess for $q(X^n)$. It is at the end of the day the denominator term of the Bayes' theorem, so it must be a valid PDF. But let's prove that anyway(since integration is therapeutic)

$$\int Z(x_1, \ldots, x_n)(dx_1\ \ldots dx_n) = \int \int f(x_1, \ldots, x_n, \overline{w})\varphi_1(\overline{w})\ d\overline{w}\ (dx_1 \ldots dx_n), \text{ where } f(x_1, x_2, \ldots, x_n, \overline{w}) = \prod_{i=1}^{n} p(x_i|\overline{w})$$

Assuming that $f(\overline{x}, \overline{w})$ is "well-behaved", we can swap the order of integration, and it immediately follows that $Z(X^n)$ is a valid PDF.

$$\int \varphi_1(\overline{w}) \left(\int f(x_1, \ldots, x_n, \overline{w})\ (dx_1 \ldots dx_n) \right)\ d\overline{w}$$

The blue portion of the integral is just 1, and therefore the entire integral simplifies to 1.
We therefore get:

$$\boxed{\int Z(X^n) d\overline{x} = 1}$$

5.1 $\mathbb{E}[\mathbf{F_n}] - \mathbf{nS}$

$$F_n = -\ln Z(x_1, x_2, \ldots, x_n)$$

1. **Mindset 1:** Bash the math

$$\mathbb{E}[F_n] - nS = \int -\ln Z(\overline{x})q(\overline{x})d\overline{x} + n \int q(x) \ln q(x) dx$$

$$\mathbb{E}[F_n] - nS = \int -\ln Z(\overline{x})q(\overline{x})d\overline{x} + \int q(\overline{x}) \ln q(\overline{x}) d\overline{x}$$

$$\mathbb{E}[F_n] - nS = -\int q(\overline{x}) \ln \left(\frac{Z(\overline{x})}{q(\overline{x})} \right) d\overline{x}$$

$$\boxed{\mathbb{E}[F_n] - nS = D_{KL}(q(\overline{x})||Z(\overline{x}))}$$

2. **Mindset 2:** Let's observe...

$$\mathbb{E}[F_n] - nS = \mathbb{E}[F_n - nS_n]$$

What is $\mathbf{F_n} - \mathbf{nS_n}$?

$$F_n - nS_n = -\ln Z(\overline{x}) + \sum_{i=0}^{n} \ln q(x_i)$$

$$F_n - nS_n = -\ln Z(\overline{x}) + \ln \left(\prod_{i=0}^{n} q(x_i) \right)$$

Since the observations are all independent,

$$F_n - nS_n = -\ln Z(\overline{x}) + \ln q(\overline{x}) = -\ln\left(\frac{Z(\overline{x})}{q(\overline{x})}\right)$$

And hence we get:

$$\boxed{\mathbb{E}[F_n] - nS = D_{KL}(q(\overline{x})||Z(\overline{x}))}$$

5.2 Difference between $\mathbb{E}[G_n - S]$ and $\mathbb{E}[F_n - nS_n]$

1. The first is the D_{KL} of the truth over the *predictive*, whilst the second is the is the D_{KL} of the truth over the *marginal likelihood*.

2. The difference between these two quantities loosely tells us how much the model has improved from before the bayesian update to after.

End of Q5

§6 Summing Up

$$\boxed{\text{I am not re-defining the definitions in the original task document}}$$

6.1 Finding the ML estimate: $\hat{a}$

Say we observe n tosses of the coin, and we record our observations in $\overline{x} = (x_1, x_2, \ldots, x_n)$. Then our likelihood function is:

$$L(\overline{x}; a) = a^{n_1}(1-a)^{n_2}$$

and our log-likelihood is:

$$l(\overline{x}; a) = n_1 \ln(a) + n_2 \ln(1-a)$$

Thus, we find our ML estimate by differentiation:

$$\left(\frac{\partial l}{\partial a}\right) = \frac{n_1}{a} - \frac{n_2}{1-a} \overset{set}{=} 0$$

Solving this for a, and fancifully calling the solution $\hat{a}$, we get the ML estimate:

$$\boxed{\hat{a} = \frac{n_1}{n}}$$

And so, we also get:

$$\boxed{\hat{p}(x|\hat{a}) = \begin{cases} \frac{n_1}{n} & \text{if } x = 1, \\ \frac{n_2}{n} & \text{if } x = 0, \end{cases}}$$

6.2 Finding the Bayesian Predictive

- **Model:** $p(x|a) = a^x(1-a)^{1-x}$
- **Prior:** $\varphi_1(a) = 1, 0 \leq a \leq 1$
- **Posterior:**

$$\varphi_2(a) = \frac{\left(\prod_{i=1}^{n} a^{x_i}(1-a)^{1-x_i}\right) \varphi_1(a)}{\int_0^1 \left(\prod_{i=1}^{n} a^{x_i}(1-a)^{1-x_i}\right) \varphi_1(a)\ da}$$

$$\varphi_2(a) = \frac{a^{n_1}(1-a)^{n_2}}{\int_0^1 a^{n_1}(1-a)^{n_2}\ da}$$

$$\boxed{\varphi_2(a) = \frac{a^{n_1}(1-a)^{n_2}}{\beta(n_1+1, n_2+1)}}$$

- **Predictive density:**

$$\hat{p}(x|X^n) = \int_0^1 a^x(1-a)^{1-x}\varphi_2(a)\ da$$

$$\hat{p}(x|X^n) = \frac{1}{\beta(n_1+1, n_2+1)} \int_0^1 \left(a^x(1-a)^{1-x}\right)\left(a^{n_1}(1-a)^{n_2}\right)\ da$$

$$\hat{p}(x|X^n) = \frac{\beta(x+n_1+1, 2+n_2-x)}{\beta(n_1+1, n_2+1)}$$

Since $x, n_1, n_2 \in \mathbb{N}$, we can expand the above expression as a factorial using the gamma function.

$$\hat{p}(x|X^n) = \frac{(x+n_1)!(1+n_2-x)!(n+1)!}{(2+n)!(n_1)!(n_2)!}$$

Therefore we finally get:

$$\boxed{\hat{p}(x|X^n) = \begin{cases} \frac{n_1+1}{n+2} & \text{if } x = 1, \\ \\ \frac{n_2+1}{n+2} & \text{if } x = 0 \end{cases}}$$

Note: This also **salvages** the given expression in the PODASIP question.

6.3 Finding the $\mathbf{D_{KL}}$ expressions

In the discrete case, we have the following expressions for G_n, S and D_{KL}:

$$S = -\sum_x q(x) \ln q(x)$$

$$G_n = -\sum_x q(x) \ln \hat{p}(x)$$

$$D_{KL} = \sum_x q(x) \ln\left(\frac{q(x)}{\hat{p}(x)}\right)$$

In this case, x runs over the set $0, 1$.
Let $K = a_0 \ln(a_0) + (1-a_0)\ln(1-a_0)$

- **Maximum Likelihood Estimate:**

$$K_{ML} = K - \left(a_0 \ln\left(\frac{n_1+\epsilon}{n+2\epsilon}\right) + (1-a_0)\ln\left(\frac{n_2+\epsilon}{n+2\epsilon}\right)\right)$$

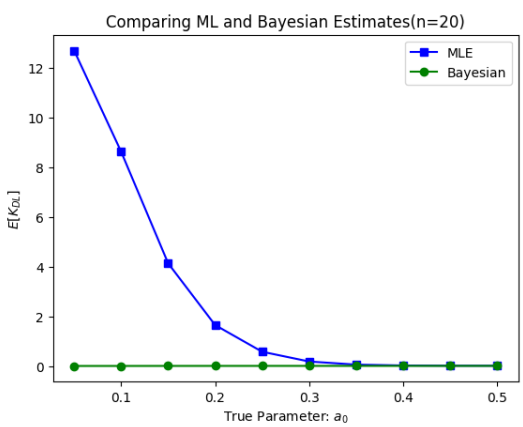
- **Bayesian Predictive:**

$$K_{BP} = K - \left(a_0 \ln\left(\frac{n_1+1}{n+2}\right) + (1-a_0)\ln\left(\frac{n_2+1}{n+2}\right)\right)$$

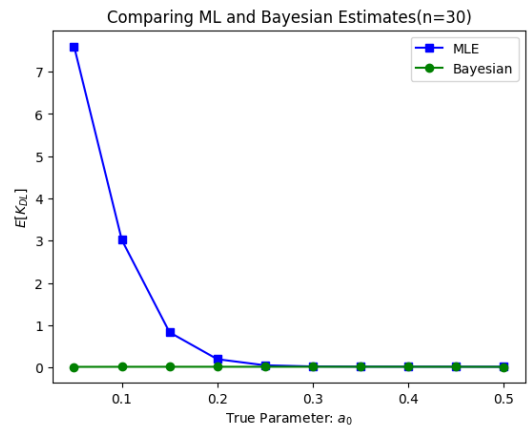
6.4 Yay some code (finally)!

- I've typed my code in Google Colab, please click here to find my notebook.
- I plotted the beta prior graphs on desmos, please click here to find the graphs.

The following are the plots I got for $n = 20, 30, 100, 500$ and $n = 1000$:

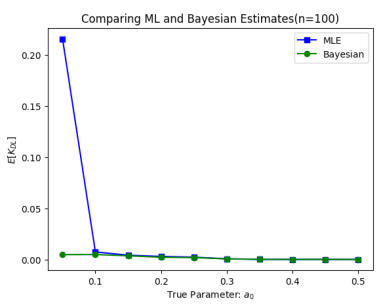


(a) n = 20

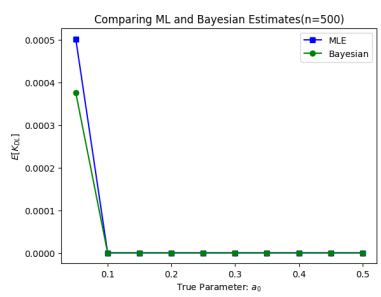


(b) n = 30

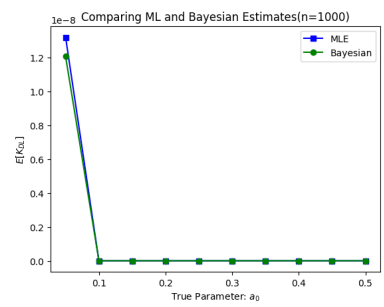
Figure 1: Relatively smaller number of samples



(a) n = 100



(b) n = 500



(c) n = 1000

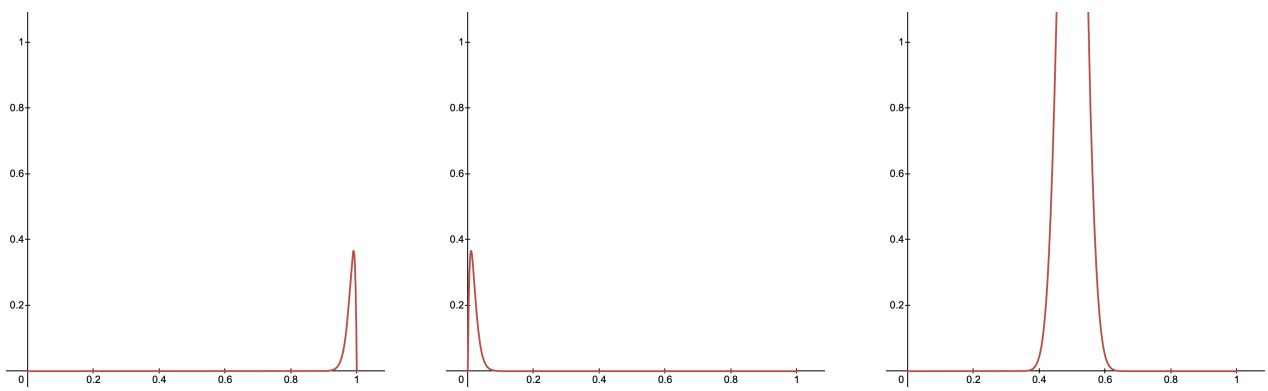
Figure 2: Relatively larger number of samples

Observations:

1. As $a_0 \to 0.5$(the most uncertain coin toss), both estimates seem to yield similar results
2. At all n values, the $\mathbb{E}[D_{KL}]$ for the Bayesian predictive is always smaller than that for the ML estimate, especially when a_0 is smaller.
3. As $n \to \infty$, we can see $WLLN$ at play; both estimates become nearly identical!

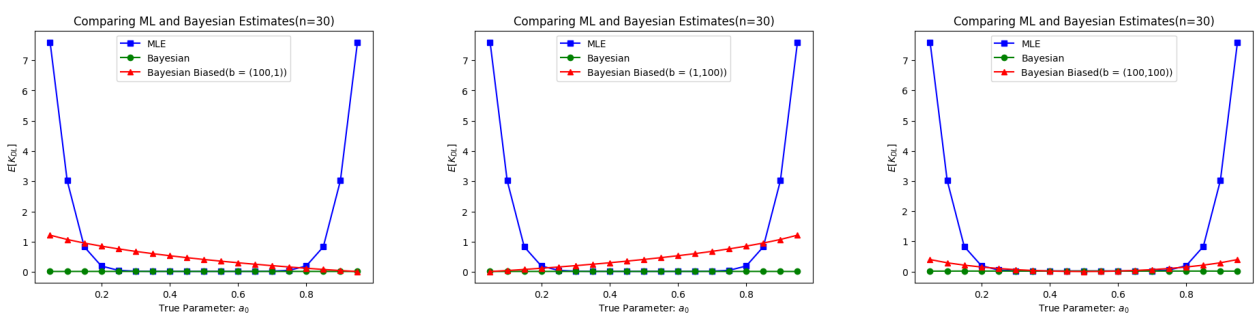
Learnings:

1. The biggest point of difference is at the extremes, at $a_0 \to 0$ and by symmetry also at $a_0 \to 1$. This is a fundamental drawback of the MLE method. Essentially, the MLE method considers only the first moment: the mean, whilst the Bayesian approach "considers" all possible moments through the integration.
2. The fact that D_{KL} doesn't blow up shows us that the Bayesian predictive accounts better for the extreme cases.
3. Suppose I change the prior from $\varphi_1(a) = 1$ to $\varphi_1(a) = c_0 a^{b_1}(1-a)^{b_2}; 0 \leq a \leq 1$ (b_1 and b_2 are biases in our prior belief about a) unless this is actually true, this will just make our results terrible, especially for smaller n values.



(a) Right biased (b) Left biased (c) Center biased

Figure 3: Graphs of a biased beta prior: $\varphi_1(a) = c_0 a^{b_1}(1-a)^{b_2}$. The center biased beta prior is SO biased at the center it just clipped the available space on this page!



(a) Right biased (b) Left biased (c) Center biased

Figure 4: $E[D_{KL}]$ with the biased prior

4. In the limit $n \to \infty$, both basically become the same. Therefore, Frequentists and Bayesian statisticians should not fight - in theory both methods work.

End of Q6